

Topic to discuss

- Runge-Kutta 4th order
- Numerical Problem
- Homework Problem

Runge-Kutta fourth Order / RK4 Method

It is one of the most popular and accurate methods because it uses four slope values to calculate the next value of y .

In Euler method, we use only one slope.

In RK2 method, we use two slopes.

But in RK4 method, we calculate four slopes:

k_1, k_2, k_3, k_4

Then we take a weighted average of these four slopes to get the next value of y .

RK4 Formula :

$$\text{Let } \frac{dy}{dx} = f(x, y)$$

$$k_1 = h f(x_n, y_n)$$

$$k_2 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_n + h, y_n + k_3)$$

$$\text{Then, } y_{n+1} = y_n + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

Given, x_0, y_0 and $f(x, y)$

Considered 'h' if not given

x_0 = Given

$x_1 = x_0 + h \rightarrow$ find y_1

$x_2 = x_1 + h \rightarrow$ find y_2

$x_3 = x_2 + h \rightarrow$ find y_3

$x_4 = x_3 + h \rightarrow$ find y_4

1st Iteration: $n=0$

$$k_1 = h f(x_0, y_0)$$

$$k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_0 + h, y_0 + k_3)$$

$$y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

2nd Iteration: $n=1$

$$k_1 = h f(x_1, y_1)$$

$$k_2 = h f\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_1 + h, y_1 + k_3)$$

$$y_2 = y_1 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

3rd Iteration: $n=2$

$$k_1 = h f(x_2, y_2)$$

$$k_2 = h f\left(x_2 + \frac{h}{2}, y_2 + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_2 + \frac{h}{2}, y_2 + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_2 + h, y_2 + k_3)$$

$$y_3 = y_2 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

4th Iteration: $n=3$

$$k_1 = h f(x_3, y_3)$$

$$k_2 = h f\left(x_3 + \frac{h}{2}, y_3 + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_3 + \frac{h}{2}, y_3 + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_3 + h, y_3 + k_3)$$

$$y_4 = y_3 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Numerical Problem :

Q: Compute $y(0.8)$ by Runge-Kutta Method correct up to five decimal places from the equation $\frac{dy}{dx} = xy$

$y(0) = 2$, taking $h = 0.2$

Solution:

$$\text{Given, } \frac{dy}{dx} = f(x, y) = xy$$

$$x_0 = 0$$

$$y_0 = 2$$

$$h = 0.2$$

Using Runge-Kutta Method of 4th order,

1st approximation: Finding y_1 ,

For, $n=0$,

$$f(x_0, y_0) = x_0 y_0 = 0 \times 2 = 0$$

$$\Rightarrow K_1 = h f(x_0, y_0) = 0.2 \times 0 = 0$$

$$\begin{aligned}\Rightarrow K_2 &= h f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right) \\ &= 0.2 f\left(0 + \frac{0.2}{2}, 2 + \frac{0}{2}\right) \\ &= 0.2 f(0.1, 2) = 0.2 [0.1 \times 2] = 0.04\end{aligned}$$

$$\begin{aligned}\Rightarrow K_3 &= h f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right) \\ &= 0.2 f\left(0 + \frac{0.2}{2}, 2 + \frac{0.04}{2}\right) \\ &= 0.2 f(0.1, 2.02) = 0.2 [0.1 \times 2.02] \\ &= 0.0404\end{aligned}$$

$$\begin{aligned}
 \Rightarrow K_4 &= hf(x_0 + h, y_0 + K_3) \\
 &= 0.2 f(0 + 0.2, 2 + 0.0404) \\
 &= 0.2 f(0.2, 2.0404) \\
 &= 0.2 [0.2 \times 2.0404] \\
 &= 0.081616
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, } y_1 &= y_0 + \frac{1}{6} (K_1 + 2K_2 + 2K_3 + K_4) \\
 &= 2 + \frac{1}{6} (0 + 2 \times 0.04 + 2 \times 0.0404 + 0.081616) \\
 &= 2.040402667
 \end{aligned}$$

$$\text{So, } y(0.2) = 2.040402667$$

Second Approximation:

Finding y_2

$$x_1 = x_0 + h = 0 + 0.2 = 0.2$$

$$y_1 = 2.040402667$$

for $n=1$,

$$f(x_1, y_1) = x_1 y_1 = 0.2 \times 2.040402667 = 0.408080533$$

Now calculating K_1, K_2, K_3, K_4 & y_2

$$K_1 = h f(x_1, y_1) = 0.2 \times 0.408080533 = 0.0816161$$

$$K_2 = h f\left(x_1 + \frac{h}{2}, y_1 + \frac{K_1}{2}\right)$$
$$= 0.2 f\left(0.2 + \frac{0.2}{2}, 2.040402667 + \frac{0.0816161}{2}\right)$$

$$= 0.2 f(0.3, 2.08121072)$$

$$= 0.2 [0.3 \times 2.08121072]$$

$$= 0.124872643$$

$$K_3 = h f \left(x_1 + \frac{h}{2}, y_1 + \frac{K_2}{2} \right)$$

$$= 0.2 f \left(0.2 + \frac{0.2}{2}, 2.04040266 + \frac{0.124872643}{2} \right)$$

$$= 0.2 f(0.3, 2.10283998)$$

$$= 0.2 [0.3 \times 2.10283998]$$

$$= 0.1261703993$$

$$K_4 = h f(x_1 + h, y_1 + K_3)$$

$$= 0.2 f(0.2 + 0.2, 2.04040266 + 0.1261703993)$$

$$= 0.2 f(0.4, 2.166573066)$$

$$= 0.2 [0.4 \times 0.173325845]$$

$$= 0.17332584$$

$$\begin{aligned} \text{So, } y_2 &= y_1 + \frac{1}{6} (K_1 + 2K_2 + 2K_3 + K_4) \\ &= 2.04040266 + \frac{1}{6} [0.0816161 + 2(0.124872643) \\ &\quad + 2(0.1261703993) + 0.17332584] \end{aligned}$$

$$y_2 = 2.1665739$$

$$\text{So, } y_2 = y(x_2) = y(0.4) = 2.1665739$$

3rd Approximation:

finding y_3 ,

$$x_2 = x_1 + h = 0.2 + 0.2 = 0.4$$

$$y_2 = 2.1665739$$

Now calculating K_1, K_2, K_3, K_4 & y_3

For $n=2$,

$$\begin{aligned}k_1 &= hf(x_2, y_2) = 0.2f(0.4, 2.1665739) \\&= 0.2[0.4 \times 2.1665739] \\&= 0.17332591\end{aligned}$$

$$\begin{aligned}k_2 &= hf\left(x_2 + \frac{h}{2}, y_2 + \frac{k_1}{2}\right) \\&= 0.2f\left(0.4 + \frac{0.2}{2}, 2.1665739 + \frac{0.17332591}{2}\right) \\&= 0.2f(0.5, 2.25323692) \\&= 0.2[0.5 \times 2.25323692] \\&= 0.2253236\end{aligned}$$

$$\begin{aligned}k_3 &= hf\left(x_2 + \frac{h}{2}, y_2 + \frac{k_2}{2}\right) \\&= 0.2f\left(0.4 + \frac{0.2}{2}, 2.1665739 + \frac{0.2253236}{2}\right) \\&= 0.2f(0.5, 2.27923581) \\&= 0.2[0.5 \times 2.27923581] \\&= 0.22792358\end{aligned}$$

$$\begin{aligned}
 k_4 &= h f(x_2 + h, y_2 + k_3) \\
 &= 0.2 f(0.4 + 0.2, 2.16657396 + 0.22792358) \\
 &= 0.2 f(0.6, 2.39449754) \\
 &= 0.2 [0.6 \times 2.39449754] \\
 &= 0.2873397056
 \end{aligned}$$

$$\begin{aligned}
 \text{So, } y_3 &= y_2 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \\
 &= 2.16657396 + \frac{1}{6} [0.17332591 + 2(0.2253236) + \\
 &\quad 2(0.22792358) + 0.2873397056] \\
 &= 2.39424247
 \end{aligned}$$

$$\text{So, } y_3 = y(x_3) = y(0.6) = 2.39424247$$

Forth Approximation

Finding y_4 ,

$$\text{for } n=3, \quad x_3 = 0.6, \quad y_3 = 2.39424247$$

Calculating k_1, k_2, k_3, k_4 & y_4

$$k_1 = h f(x_3, y_3) = 0.2 [0.6 \times 2.39424247] = 0.28730909$$

$$k_2 = h f\left(x_3 + \frac{h}{2}, y_3 + \frac{k_1}{2}\right)$$

$$= 0.2 f\left(0.6 + \frac{0.2}{2}, 2.39424247 + \frac{0.28730909}{2}\right)$$

$$= 0.2 f(0.7, 2.53789702)$$

$$= 0.2 [0.7 \times 2.53789702]$$

$$= 0.35530558$$

$$K_3 = h f \left(x_3 + \frac{h}{2}, y_3 + \frac{k_2}{2} \right)$$

$$= 0.2 f \left(0.6 + \frac{0.2}{2}, 2.39424247 + \frac{0.35530558}{2} \right)$$

$$= 0.2 f(0.7, 2.571895)$$

$$= 0.2 (0.7 \times 2.571895)$$

$$= 0.360065337$$

$$K_4 = h f(x_3 + h, y_3 + k_3)$$

$$= 0.2 f(0.6 + 0.2, 2.39424247 + 0.3600653)$$

$$= 0.2 f(0.8, 2.75430781)$$

$$= 0.2 (0.8 \times 2.75430781)$$

$$= 0.44068925$$

$$\begin{aligned}
 \text{So, } y_4 &= y_3 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \\
 &= 2.39424247 + \frac{1}{6} \left[0.28730909 + 2(0.35530558) \right. \\
 &\quad \left. + 2(0.360065337) + 0.44068925 \right] \\
 &= 2.7540309
 \end{aligned}$$

$$\text{So, } y_4 = y(x_4) = y(0.8) = 2.7540309$$

$$\text{So, } y(0.8) = 2.75403 \text{ (correct up to 5 decimal places)}$$

Homework Problem :

Q: Solve the equation $\frac{dy}{dx} = \frac{1}{x+y}$, $y(0) = 1$

for $y(0.1)$ and $y(0.2)$ using Runge-Kutta Method of 4th order.

Solution,

Given, $\frac{dy}{dx} = f(x, y) = \frac{1}{x+y}$

$$x_0 = 0$$

$$y_0 = 1$$

$$\text{Let, } h = 0.1$$

First Approximation :

Finding y_1 ,

Taking $x=0$, $x_0=0$, $y_0=1$, $h=0.1$, $f(x,y) = \frac{1}{x+y}$

Calculating k_1, k_2, k_3, k_4 & y_1

$$k_1 = h f(x_0, y_0) = 0.1 f(0, 1) = 0.1 \left[\frac{1}{0+1} \right] = 0.1$$

$$k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$= 0.1 f\left(0 + \frac{0.1}{2}, 1 + \frac{0.1}{2}\right)$$

$$= 0.1 f(0.05, 1.05)$$

$$= 0.1 \left[\frac{1}{0.05+1.05} \right] = 0.090909$$

$$K_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right)$$

$$= 0.1 f\left(0 + \frac{0.1}{2}, 1 + \frac{0.090909}{2}\right)$$

$$= 0.1 f(0.05, 1.0454545)$$

$$= 0.1 \left[\frac{1}{0.05 + 1.0454545} \right]$$

$$= 0.091286307$$

$$K_4 = h f(x_0 + h, y_0 + K_3)$$

$$= 0.1 f(0 + 0.1, 1 + 0.091286307)$$

$$= 0.1 f(0.1, 1.091286307)$$

$$= 0.1 \left[\frac{1}{0.1 + 1.091286307} \right]$$

$$= 0.08394287$$

$$\begin{aligned}
 \text{So, } y_1 &= y_0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \\
 &= 1 + \frac{1}{6} [0.1 + 2(0.090909) + 2(0.091286307) \\
 &\quad + 0.08394287) \\
 &= 1.091388945
 \end{aligned}$$

$$\text{So, } y_1 = y(x_1) = y(0.1) = 1.091388945$$

Ans.

Second Approximation :

Finding y_2 ,

Taking $n=1$

We have, $x_1 = 0.1$, $y_1 = 1.091388945$

Calculating, K_1, K_2, K_3, K_4 & y_2

$$K_1 = h f(x_1, y_1)$$

$$= 0.1 f(0.1, 1.091388945)$$

$$= 0.1 \left[\frac{1}{0.1 + 1.091388945} \right]$$

$$= 0.0839364$$

$$K_2 = h f \left(x_1 + \frac{h}{2}, y_1 + \frac{K_1}{2} \right)$$

$$= 0.1 f \left(0.1 + \frac{0.1}{2}, 1.091388945 + \frac{0.0839364}{2} \right)$$

$$= 0.1 f(0.15, 1.13335731)$$

$$= 0.1 \left[\frac{1}{0.15 + 1.13335731} \right]$$

$$= 0.07792146$$

$$K_3 = h f \left(x_1 + \frac{h}{2}, y_1 + \frac{K_2}{2} \right)$$

$$= 0.1 f \left(0.1 + \frac{0.1}{2}, 1.091388945 + \frac{0.07792146}{2} \right)$$

$$= 0.1 f(0.15, 1.13034982)$$

$$= 0.1 \left[\frac{1}{0.15 + 1.13034982} \right]$$

$$= 0.078104165$$

$$\begin{aligned}
 K_4 &= h f(x_1 + h, y_1 + K_3) \\
 &= 0.1 f(0.1 + 0.1, 1.091388945 + 0.078104165) \\
 &= 0.1 [0.2, 1.1694932565] \\
 &= 0.073020455
 \end{aligned}$$

$$\begin{aligned}
 \text{So, } y_2 &= y_1 + \frac{1}{6} (K_1 + 2K_2 + 2K_3 + K_4) \\
 &= 1.1695569519
 \end{aligned}$$

$$\text{So, } y_2 = y(x_2) = y(0.2) = 1.1695569519$$

Ans.

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